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DEVELOPMENT OF AUTOMATIC PACKAGING SYSTEM USING PLC AND SCADA FOR INDUSTRIES

Tarun Gupta
Student, EIED, Thapar Institute of Engineering and Technology, Patiala,

Sangeeta Kamboj
Assistant Professor, EIED, Thapar Institute of Engineering and Technology, Patiala

ABSTRACT

The application of Programmable Logic Controller (PLC) and Supervisory Control and Data Acquisition (SCADA) is widely known for industrial application to control a mechanical movement for an efficient production. The paper presents designing of PLC and SCADA based system to automate packaging section in industries. The packaging of the final made product which is ready to be packed in the boxes/cartons for further delivery needs to be automated to reduce the man work and packaging time and at the same time to fasten the packaging speed. The simulation studies of the designed system for various sections such as Bottle filling process, Case erector, Filing, Taping, Weighing , Labelling, Palletizer, Wrapping and Loading are also presented in the paper. The paper also presents analysis of manual packaging system of Verka Milk Plant, Mohali, Punjab, India to find engineering problem and opportunities. It is found that designed system gives advantages of performing multiple processes on a single screen using single PLC and can be visualized on a single screen using SCADA.

Key words: PLC, SCADA, Smart Parking System, TIA Portal, Sensor.

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1. INTRODUCTION

Today's extremely increasing level of competitiveness within the industries demands prime quality and most consistent products at lowest price. To meet this challenge many industries are considering industrial automation. The term automation means the use of various control systems for operating the equipments like processes in factories, industrial machinery and processes and other applications with reduced or very less human intervention [1]. In general industrial automation is defined as the use of various set technologies and automatic control

devices combination of which results the automatic operation and control of industrial processes without any significant human intervention and with a purpose of achieving superior performance better than manual control [2]. In the most of industries all the packaging processes i.e. case erector, filling, taping, weighing, labeling, palletizing process and loading of packaging system are still manual and a team of workers must be present in the industry to perform the packaging work. The proposed work introduces the automatic packaging system that could be incorporated in these industries so as to make the packaging process automatic, error free and fast [3].

The paper proposed a system to make or modify the present manual packaging system into automatic packaging system. In the present work system for automation of packaging section is designed using PLC and SCADA for industries. The designed system gives advantages of performing multiple processes on a single screen and using a single PLC. Complete packaging process is included and visualized on a single screen using SCADA. The programming for the system has been developed and done on the PLC S7-300 through the use of Totally Integrated Portal (TIAportal). The SCADA system is used to control and monitor the performance of the designed system using Siemens software WinCC RT Professional.

2. MANUAL PACKAGING SYSTEM

Verka milk plant, Mohali, Punjab has been visited to analyze the manual packaging system of the plant to find engineering problem and opportunities. The working of the manual packaging system used at the plant is explained as under:

- When the production of product (say) milk gets completed through a process flow; milk gets packed in the bottles of different sizes through an automatic system; after that when these packets have to be packed in the cartons/boxes, then an manual team of 15-20 workers has to be present there at the plant to fill these packets in boxes; further to send these boxes at desired places.
- Fig.1 shows that a manual team have the responsibility of making the initial flat box to open box by opening the box.
- As shown in Fig.2, a few workers have the responsibility of filling the ready bottles/ product in those boxes.
- A few workers have the responsibility of taping as can be seen in Fig.3, weighing, labeling and sealing this box to further load these packed boxes into trucks as shown in Fig.4.

Figure 1 Box erection by manual team

Figure 2 Manual box filling

Figure 3 Manual box taping

Figure 4 Manual box loading

It can be observed that manual packaging system have their own drawback of less accuracy, slow implementation and likely human errors as shown in above pictorial presentations.

3. DESIGN METHODOLOGY

The flow chart of proposed work has been depicted in Fig.5 to implement the designed system. There are many sections which need to be automated for automation of the packaging unit of industries. In one of the sections i.e. weighing section, weight of final made product is taken and check whether it exceeds the predefined limit or not. If it exceeds, product is rejected otherwise given to labeling section for automation of the complete process.

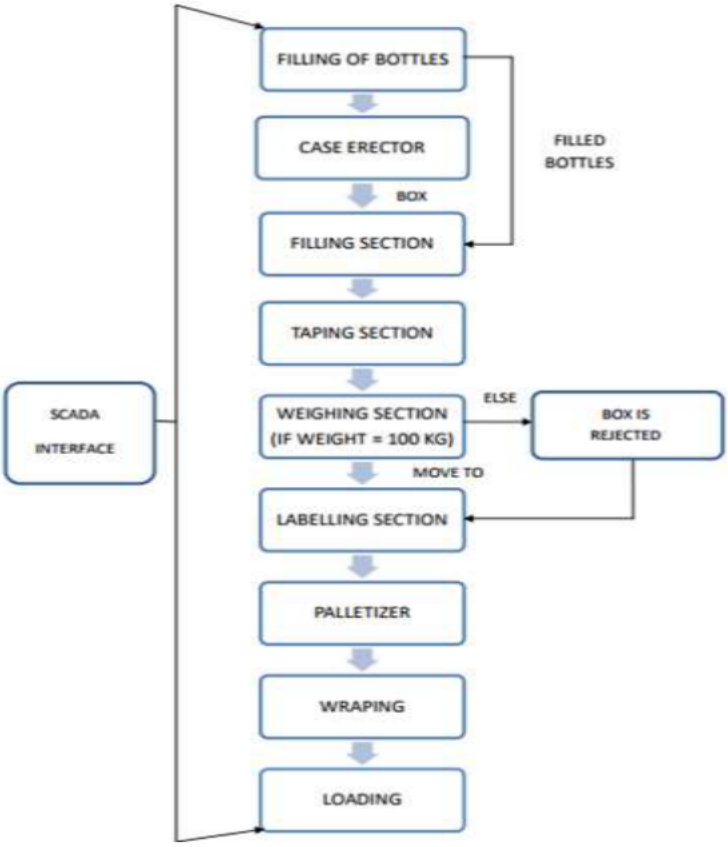

Figure 5 Flow chart of system proposed

The programming for the whole project is to done on the PLC S7-300 through the use of Totally Integrated Portal (TIA portal). The PLC S7-300 has been chosen because of the availability of simulation in this PLC which makes it easier to monitor various parameters involved in the project. Ladder logic takes in account the programming functions that are graphically displayed to resemble the symbols that are used in hard wired control diagrams. And for the implementation of the system 11 networks were made using counters, comparators, move instructions, input and output coils etc.

The SCADA system has been used to control and monitor the performance of the designed system and WinCC software is used for this purpose of visualization. The following steps need to be followed for establishing the connection between the SCADA system and the PLC:

- In the TIA portal, start new project and configure PLC S7-300.
- From the ADD NEW DEVICE option, add SIMATIC HMI screen, i.e. Win-CC RT professional which is basically the SCADA system.
- The IP addresses of the PLC and WinCC professional must be different. However, the IP addresses of the WinCC professional and the computer must be same.
- While coding the PLC each element and memory is assigned a tag name which is associated with different properties/events of HMI screen. All the PLC tags can be edited and added in show all PLC tags option as shown in Fig.6.

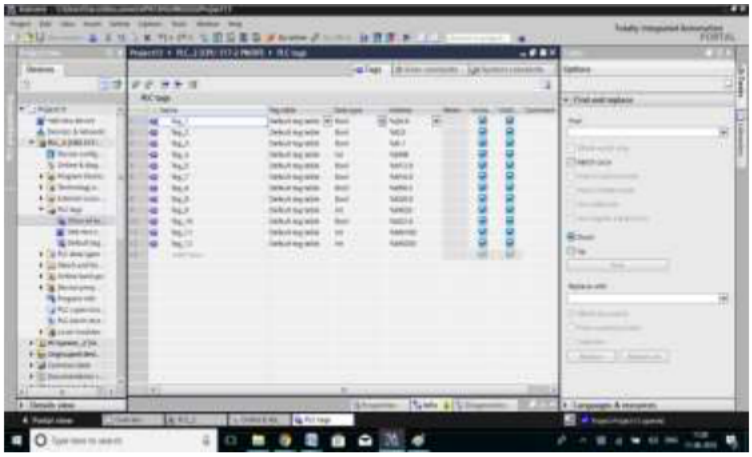

Figure 6 PLC tags

As shown in Fig.7, I0.0, Q0.0 and M1.0 addresses are of a single byte and MW8 and MD12 are of data type word (2 byte) and double word type (4 byte).

After the successful connection establishment of SCADA system and PLC, simulation can be run and the design can be easily monitored and controlled as shown in Fig.8.

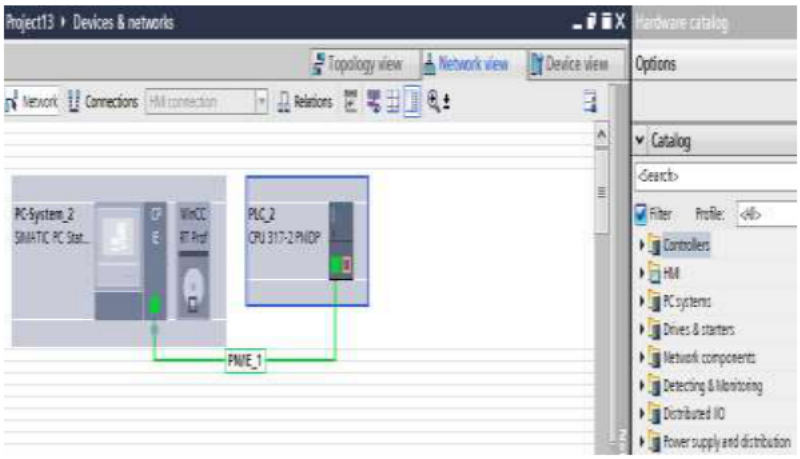

Figure 7 Creating connection between PLC and HMI

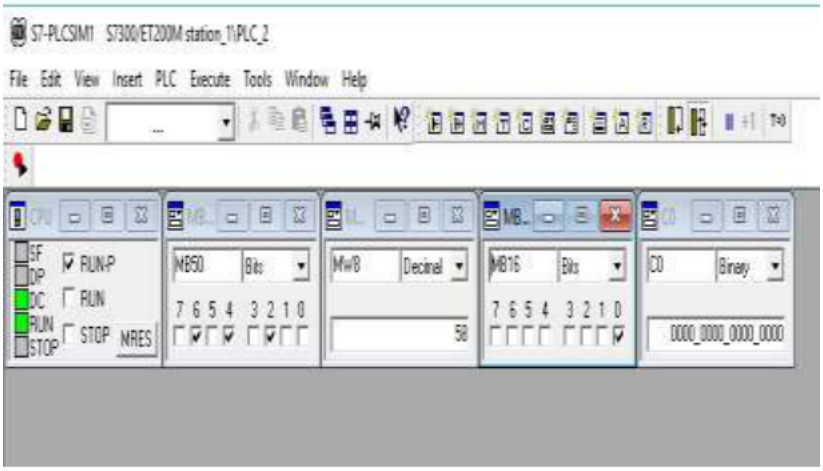

Figure 8 PLC Simulation

For designing the screen in "HMI screen view" following properties are associated with tags to carry out the complete process as can be seen in Fig.4.

- Animation: To provide any sort of movement to objects, tags are associated in animation by giving a range of value from starting coordinates to some final coordinates.
- Events: To associate different tasks such as On Click, On Right Click, Activate, Release keyboard, etc with functions like Setbit, Reset bit, Show logon screen etc process tags are attached in events.
- Text and graphics list: Process tags are associated with text list or graphic list for different values to show animation at a static position.

4. SIMULATION STUDIES

4.1. Bottle Filling Process

Each step which is included in bottle filling i.e. cleaning, washing, liquid filling and capping are made automatic and this wholesome process is showcased on a single screen as shown in Fig.9. As shown firstly the un-cleaned bottles come from the main entry gate; then this bottles pass through a cleaner where through a combined process of washer and cleaner these bottles gets cleaned. Now these cleaned bottles further move towards main tank and valve for filling these bottles with the desired liquid. After filling process, these bottles pass through a capping machine where capping of these bottles is done. Now these capped bottles pass through an exit gate to further pack these bottles in an erected box.

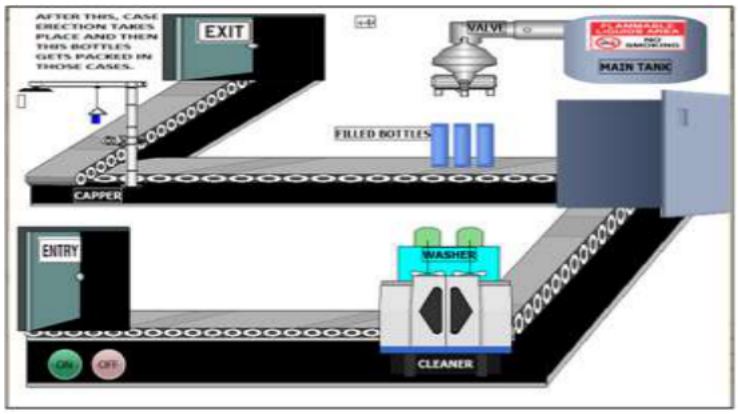

Figure 9 Automatic bottle filling

4.2. Case Erector

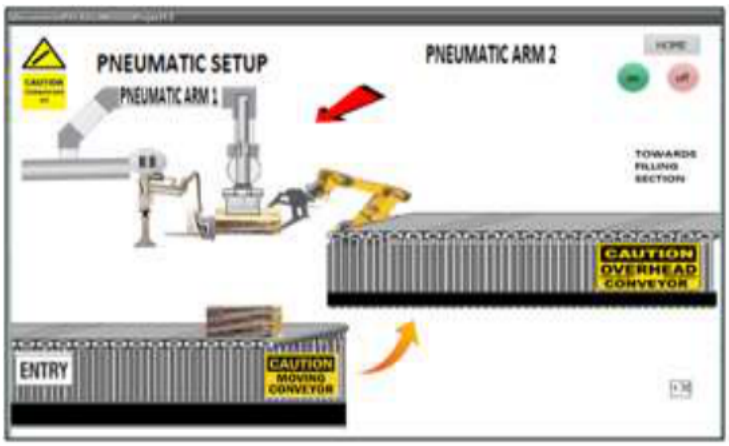

Figure 10 Automatic case erector

The conveyor combination keeps the flow of cases in-line and also provides less margin error while processing the case. This feature erects the cases in a same “upright” orientation in which they will be packed. This section of the process uses pneumatic setup for its application. Pressure is applied on the flat box which is moving on the conveyor and two pneumatic arms pulls the box through pressure and puts the box into the second conveyor for further filling of the bottles into these erected boxes as shown in Fig.10.

4.3. Filling Section

In Fig.11 of the process the filled bottles ready for packaging are packed in these erected boxes. Each carton is packed with the pack of 6 bottles through this automated process as shown in below picture. One conveyer is bringing the open carton box (as illustrated in case erector section) and the other conveyer is taking the load of filled bottles over it (which is filled in the bottle filling section). In this section the gate only gets opened when the bottle count=6 and further through an automatic counter and sensor operation a bunch of 6 bottles gets filled in the carton boxes to move ahead this box for further taping processes .

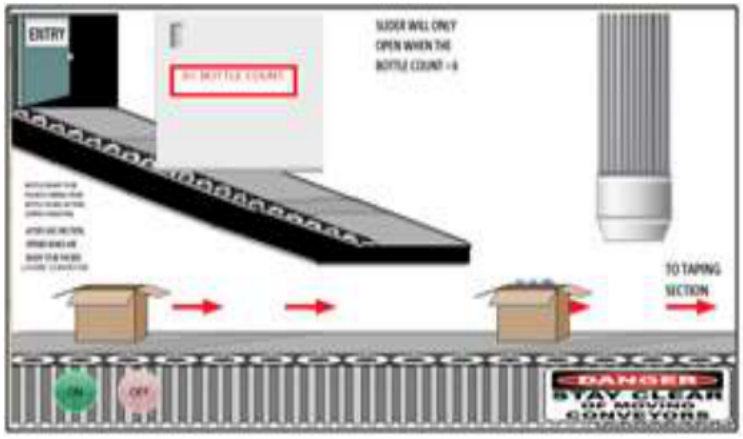

Figure 11 Automatic box filling

4.4. Taping Section

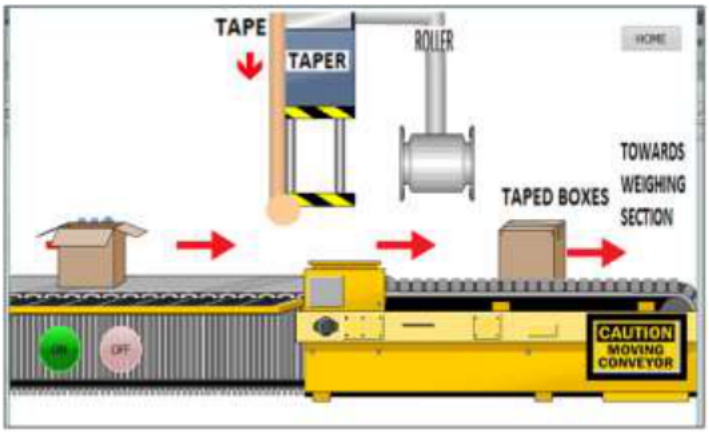

Figure 12 Automatic taping

In this section the filled boxes are taped to ensure the safety of the products inside the box. This is a single conveyor section where the filled boxes undergo through a taping machine in which a tape is hanging at the bottom of this machine and when the carton box moves through this machine, the boxes gets taped. The taping is done automatically through a PLC program

which is configured on the system. Now this taped boxes moves through a roller which moves over the taped boxes in order to ensure that the proper taping is applied on the box as shown in Fig.12.

4.5. Weighing Section

The taped boxes pass through a weighing transmitter in which weighing of boxes is done. When the weight of box is correct (by default 100 KG in this case), then the LED flashes green and this box is accepted and it moves to the conveyor for further processes. When the weight of box is incorrect (80 KG in this case), then the LED flashes red and this box is reject and it is moved away from the conveyor through an setup of pushing pistons which push away the box away from conveyor. When the transmitter is empty, then the LED flashes yellow indicating the next box to move in for weighting process as shown in Fig.13.

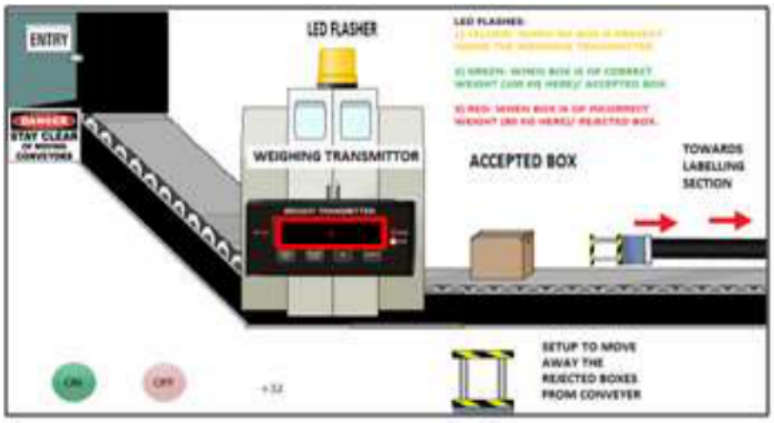

Figure 13 Automatic weight check

4.6. Labelling Section

After weighing of the boxes, now accepted boxes gets labeled the box as per the industrial specifications. In this section the label is applied on the box when it passes through a product sensor and shooter. After this the labeled box passes through a heat tunnel; a heat tunnel is an enclosed and heated area that creates an heated local environment around said object and it does not apply heat directly to that object inside it. The goal is to shrink label on the box. Now these boxes pass through a cold air cooling fan where cooling process of this heated label takes place and after this final ready labeled box is ready as shown in Fig.14 to move to other sections.

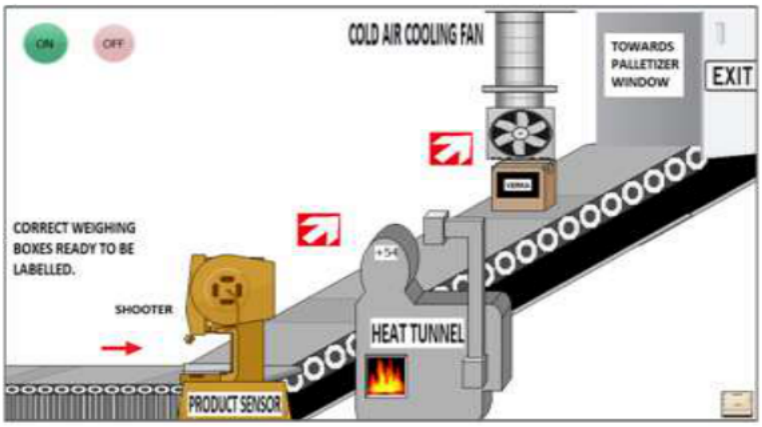

Figure 14 Automatic labelling

4.7. Palletizer Section

In this section stacking of the labeled boxes onto a pallet or stack for further storage or wrapping process takes place. In this setup developed three cranes are used; these cranes move vertically and pick up the labeled boxes and they move into the direction of trolley where these boxes are stacked onto a trolley. Here in this case a set of 12 boxes gets stacked onto the trolley. This automated section developed will make the problems of space requirements, storage and material handling limited in the industries as shown in Fig.15.

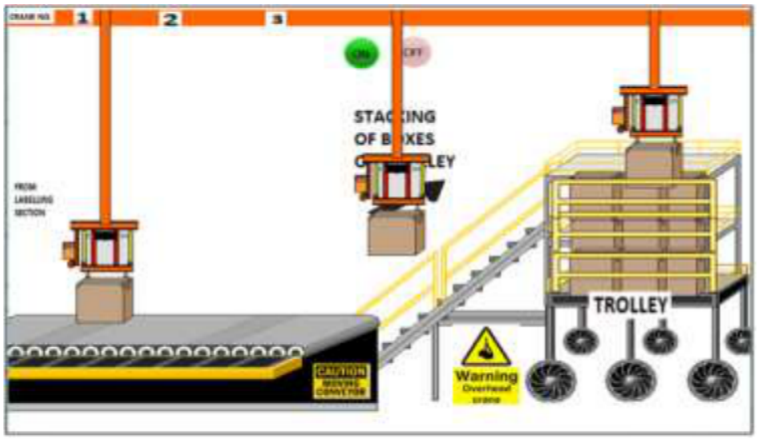

Figure 15 Automatic palletize section

4.8. Wrapping Section

In this section wrapping of stack of boxes takes place. The trolley from the palletize section that is carrying on it the stack of 12 boxes moves towards wrapping section, from where this stack of boxes moves under a wrapping machine and here the wrapping material moves over the boxes and wrapping gets completed. These wrapped box moves through an air compressor where this wrapping gets shrink and this final wrapped box is now further ready for loading as shown in Fig.16.

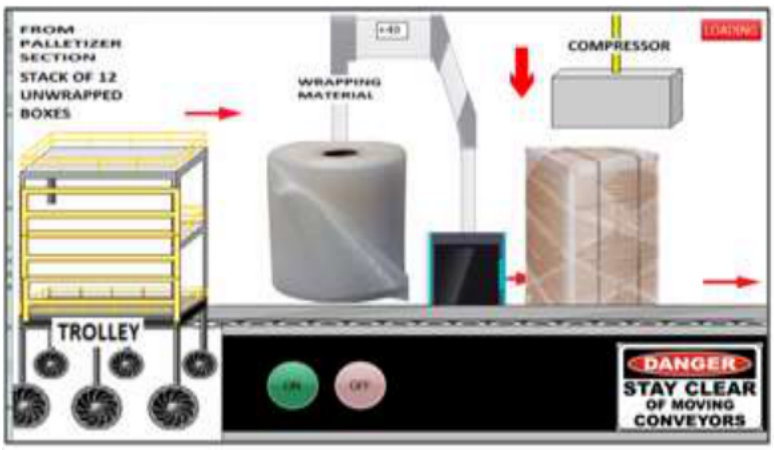

Figure 16 Automatic warping

4.9. Loading Section

At this stage packaging gets completed and now the final packed boxes are ready to be delivered to further places. So in this section loading of these boxes is done. Through an overhead conveyor setup the packed boxes are brought inside the truck. Here total of 24 (12 +

12) boxes are getting loaded in this truck as shown in Fig.17. After the loading in truck gets completed, these trucks are further delivered to their target locations.

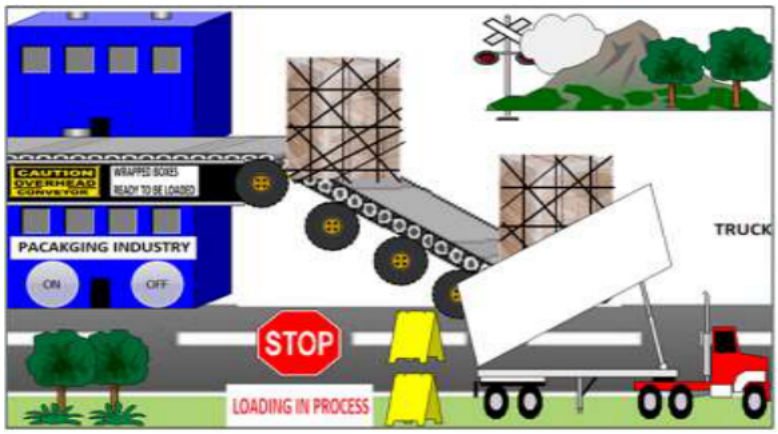

Figure 17 Automatic loading of boxes

5. CONCLUSIONS

The modeling of a automatic packaging system using PLC and SCADA has been done in the paper. The fundamental reason for PLC in automation is utilized to control the entire system and the major reason for utilizing SCADA is to monitor process and store data for future improvements. Moreover in the program developed on PLC, troubleshooting of program and making changes to become very easy due to simplicity of the program. It can be concluded that any changes may be made to the program in between the process and then these can be downloaded into load memory of CPU without any inconvenience and without putting the CPU in stop mode. The It can also be concluded that the cost of establishment of designed system would not be cheap but it can effectively keep running for a long stretch of time. In long term such automated systems can prove profitable for any industry.

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REFERENCES

[1] Algitta, A.A., Mustafa, S., Ibrahim, F., Abdalruof, N. and Yousef, M., 2015, "Automated packaging machine using PLC", IJISET-International Journal of Innovative Science, Engineering and Technology, 2(5), 282-288, 2015.

[2] Lingappa, S. and Bongale, V., Sreerajendra, "PLC Controlled Low Cost Automatic Packing Machine", International Journal of Advanced Mechanical Engineering, 4(7), 803-811, 2014.

[3] Kumbhar, V.B., Gokhale, S.D., Kumbhar, V.S. and Patil, M.K., "Implementation of Customised SCADA for Cartoner Packaging machine for Cost Effective Solution", International Research Journal of Engineering and Technology, 4(1), 2017.

[4] Kanimozh, D. and Nantheni, B., "PLC controlled automatic food packaging machine", International Journal of Engineering Trends and Technology (IJETT), 30(1), 33-36, 2015.

[5] Bagve, R., Kumbhar, V., Bhat, M.D., Verleker, S.V., Fernandes, J., Bagve, R., Kumbhar, V., Bhat, M.D., Verleker, S.V. and Fernandes, J., "Automatic Packing Machine &Material Handling Using Programmable Logic Controller (PLC)", International Journal, 2(10), 24-29, 2016.

[6] Mahalik, N.P., "Advances in packaging methods, processes and systems", in Challenges, 5(2),374-389,2014.

[7] Liu, Z., Li, M., Chen, Z., Lin, Z. and Liu, X., "The automatic packaging machine design based on reconfigurable theory," Consumer Electronics, Communications and Networks IEEE International Conference, 812-815, 2011.

[8] Siemens supply, "Step 7 ladder logic for S7300 and S7400", Reference manual SITRAIN, Germany, Fourteen Edition, 2017.

[9] Siemens supply, "Simatic S7 manuals and guides", SITRAIN, Germany, Thirteenth Edition, 2016.

[10] Simatic HMI, "operator control and monitoring system", SITRAIN, Germany, Fourteenth edition, 2017.

[11] Ankush Dharkar and Dr. P. M. Daigavane Control of Variable Frequency Drives with PLC: A Review. International Journal of Electrical Engineering & Technology, 8(1), 2017, pp. 45–51.

[12] Rathore, R. S., Dr. Sharma, A. K. and Dubey, H. K. PLC based PID Implementation i n Process Control of Temperature Flow and Level. International Journal of Advanced Research in Engineering and Technology, 6 (1), 2015, pp. 19 - 27.